

Algorithms Problem Solving

Level 3

Trim Left Trim Right Trim



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Definition

String Boundary Whitespace Removal

 Program removes unwanted spaces from the left side, right side, or both sides of a string.

 Trimming is commonly used to clean input before validation or storage.

Explanation

`TrimLeft()`

-  Searches for the first character that is not a space.
-  The string is then rebuilt beginning from that position.

`TrimRight()`

-  Searches backward for the last character that is not a space.
-  The string is shortened so trailing spaces are removed.

`Trim()`

-  Combines both operations to remove spaces from both boundaries.
-  Internal spaces between words are preserved.

Empty and All-Space Strings

-  A string containing only spaces has no non-space position.
-  This case must be handled before using the returned `npos` position.

Procedures

- 1 Implement `TrimLeft()`.
- 2 Implement `TrimRight()`.
- 3 Implement `Trim()` by combining the two functions.
- 4 Test the functions with strings containing leading and trailing spaces.

💡 Example

Input:

" Sara Alterkawi "

➡ Result

Trim Left:

" Sara Alterkawi "

➡ Result

Trim Right:

" Sara Alterkawi"

➡ Result

Trim:

"Sara Alterkawi"

Note

- ◆ ``find_first_not_of()`` and ``find_last_not_of()`` are useful for locating non-space boundaries.
- ◆ Basic implementations that search only for ``'``` do not automatically remove tabs or newlines.
- ◆ ``Trim()`` should not remove meaningful spaces inside the sentence.

⊘ Warning

- ◆ ``find_first_not_of()`` can return ``string::npos`` when the string contains only spaces.
- ◆ ``substr()`` boundaries must be calculated correctly.
- ◆ If all whitespace characters need to be handled, ``isspace()`` is more appropriate than checking only ``' '``.



Comparison

▨ `TrimLeft()` / `TrimRight()` / `Trim()`

- ✓ Focus on removing whitespace from string boundaries.
- ✓ Preserve meaningful internal spaces.
- ✓ Provide different levels of cleaning.

▤ Character Removal

- ✓ Can remove selected characters from anywhere in the string.
- ✓ May alter meaningful internal content.
- ✓ Has a different purpose from boundary trimming.

✦ Summary

- ✓ Removes unwanted spaces from the beginning or end of strings.
- ✓ Provides separate left and right trimming operations.
- ✓ Combines both operations through `Trim()`.
- ✓ Demonstrates safe boundary-based string cleaning.

Thank you 😊